

Generative-AI Exposure and Structure of Work in Sub-Saharan African Labour Markets

Abstract

This paper measures how exposed workers are to generative AI in three Sub-Saharan African economies, using household surveys that reach agricultural and own-account workers and the own-use producers outside the employment boundary. Across 24,838 workers in Nigeria, Tanzania and Uganda the mean task-exposure score is 0.072, and 0.090 weighted, on the moderate 2030 Qwen 2.5 ratings, and runs 0.045 to 0.489 across nine versions. Agricultural workers average 0.029 against 0.134 for non-agricultural own-account workers and 0.150 for non-agricultural wage employees. Inside wage work, formal employees average 0.213 against 0.104 for informal ones. Clerical support, the most exposed major occupational group at 0.615, holds 0.6 percent of the sample. Giving employed workers a high-income occupational mix, with their own occupation scores, raises exposure from 0.088 to 0.209.

Keywords: generative AI, occupational exposure, informality, structural transformation, Sub-Saharan Africa.

JEL codes: J21, J24, O33, O55.

1 Introduction

Generative-AI job exposure estimates are now available for many of the world's economies (Felten et al., 2021; Eloundou et al., 2024; Gmyrek et al., 2023), and all of them examine the people the statistical system counts as employed. The 19th International Conference of Labour Statisticians resolved that production mainly for a household's own use no longer counts as employment (International Labour Organization, 2013), and the survey data these

studies examine follow that boundary. In the three African economies examined here, own-use producers are a quarter of the workers in the household surveys.

Pizzinelli et al. (2023) score worker-level microdata from labour force and household surveys in six economies, counting formal, informal and self-employed workers alike. Cazzaniga et al. (2024) apply the same classification to 142 countries through the ILO employment data. Gmyrek et al. (2023) score 59 countries at four-digit occupation codes, eight of them low income, and cascade the result up to global and income-group aggregates, which Gmyrek et al. (2025) refine on the same harmonised microdata collection. None of the four splits wage jobs by formality or counts own-use producers. Demombynes et al. (2025) score four-digit worker data for 25 low- and middle-income countries, Tanzania among them, and find self-employed and unpaid workers more exposed than paid employees once occupation group and sector are held fixed. Gmyrek et al. (2024) split exposure by formality and employment status in Latin America at two digits. This paper supplies the decomposition for household-survey data from Sub-Saharan Africa and divides exposure between agriculture, wage work and own-account work, splits wage jobs into formal and informal, and shows how far the average moves once own-use producers are counted.

This paper asks how exposed workers in three African economies are to generative AI, how that exposure is spread across occupations and employment categories, how much of the low average is a matter of where employment sits, and whether the same structural patterns show up across a wider set of African economies.

To answer these questions I use the Living Standards Measurement Study Integrated Surveys on Agriculture (LSMS-ISA), which records the main-job occupation of working household members whether or not they work in the formal economy (World Bank, 2024). I take the three LSMS-ISA waves that carry four-digit occupation codes, for Nigeria (2023), Tanzania (2020) and Uganda (2019), link each worker’s occupation to a task-level measure of generative-AI capability through an occupation crosswalk, and harmonise the result into a sample of 24,838 workers. A fourth wave, Ethiopia (2021), records only one-digit occupation codes and enters as a supplement in Appendix C.

The evidence is descriptive and it shows that exposure is low and heavily concentrated on the moderate 2030 ratings scored by Qwen 2.5, the measure’s canonical version. The pooled mean is 0.072 on a zero-to-one scale and 0.090 on survey weights. More than half of all workers, 57.5 percent, score below 0.05, and the most exposed tenth of workers holds 43 percent of the sum of worker exposure scores. On the same measure, agricultural workers average 0.029 against 0.134 for non-agricultural own-account workers and 0.150 for non-agricultural wage employees. Clerical support work scores 0.615, nearly double the next group, and employs 0.6 percent of the sample. The occupations generative AI is rated able

to reach furthest into by 2030 are the ones almost nobody in these economies works in.

Within non-agricultural wage employment, formal workers score 0.213 and informal workers 0.104, and the gap holds in all three countries. Agriculture sorts 62 percent of the sample, and formality sorts inside the 17 percent who hold non-agricultural wage jobs. The surveys ask the formality questions of wage employees alone, which leaves own-account formality unmeasured. A cross-section of African economies shows the same structure, with one qualification the worker-level data help settle. Aggregate exposure falls as informal employment and agricultural self-employment rise and climbs with income per head. Which of the two carries the relationship depends on how aggregate exposure is measured. Under a country measure built from occupation shares and the worker-level scores estimated in this paper, the agricultural share does the work and informality adds little once agriculture and income are held fixed.

These results add to the finding that low-income economies are less exposed to AI (Pizzinelli et al., 2023; Cazzaniga et al., 2024). Once agricultural, own-account and own-use workers are in the data, the low average reflects the concentration of employment in occupations assigned low exposure. Giving these three economies the occupational mix of the United Kingdom, and keeping their own occupation scores, raises exposure from 0.088 to 0.209, with managers, professionals, technicians and clerical support carrying the whole of that rise. Measured exposure differences follow the task composition of the occupation, which is how the score is built. In these three countries very few jobs contain the tasks a model is rated able to do by 2030 under the moderate scenario.

What this paper adds is coverage and granularity as the sample holds 6,372 own-use producers, whom the employment boundary keeps out of the estimates above, and it records work status, wage-job formality and the agricultural split for each of the three countries on its own.

1.1 Related literature

Work measuring the exposure of jobs to automation and AI starts from the task framework of Autor et al. (2003) and Acemoglu and Restrepo (2019, 2022), which treats an occupation as a bundle of tasks and asks which of those tasks a technology can carry out. That tradition runs from computerisation probabilities (Frey and Osborne, 2017) to AI occupational exposure scores (Felten et al., 2021; Webb, 2020) and large-language-model exposure (Eloundou et al., 2024). I use the task-level generative-AI measure of Althoff and Reichardt (2026), which rates each O*NET task for whether a language model will be able to perform it by 2030. This paper maps an existing measure onto household-survey workers at a level of employment-

status detail that existing cross-country estimates do not provide, and it adds the own-use producers outside the conventional employment boundary. A separate literature estimates realised effects of generative AI on productivity and labour demand (Brynjolfsson et al., 2023; Noy and Zhang, 2023; Hui et al., 2023), almost all of it in high-income or online settings.

A parallel literature compares exposure across countries. Alonso et al. (2020) warn that AI may widen the gap between rich and poor economies. Pizzinelli et al. (2023); Cazzaniga et al. (2024); Gmyrek et al. (2023) find that low-income economies are less exposed, given that fewer of their workers hold the occupations AI reaches. Gmyrek et al. (2025) refine the four-digit index and report global, regional and income-group exposure through the same ILO estimation framework. All of those routes measure employment as the labour force surveys define it. The household surveys used here add the own-use producers that boundary leaves out, and report the agricultural, own-account and informal structure country by country.

Informality and structural transformation are the other body of work this paper draws on. La Porta and Shleifer (2014) document how far informal and own-account work dominates employment in low-income economies, and Gollin et al. (2014) show how much labour remains in low-productivity agriculture. I set those two facts against the exposure measure and ask how much of the low African average they account for.

2 Data and measurement

2.1 What the exposure measure measures

Throughout the paper, exposure means the importance-weighted share of an occupation’s tasks that a language model is rated able to perform. Althoff and Reichardt (2026) rate every O*NET task on a four-point capability scale and record the top two points as a yes-or-no rating. The occupation score is the mean of that rating over the occupation’s tasks, weighted by the O*NET importance weight of each task. The result runs from 0 to 1. A score of 0.20 means that tasks carrying a fifth of the occupation’s total task importance are rated automatable.

The weight is an importance weight and a task can matter a great deal to a job without filling much of the working day, and the score therefore measures no share of time. The canonical ratings are also written against a moderate 2030 scenario, which asks what a model will be able to do by 2030 under a middling rate of capability advance. The score describes the reach of the technology over the next few years, and taking it as a description of what models do at the time of writing would read it too high.

The level of exposure depends on which release of the ratings is used, and I hold the release

fixed at the version of 7 June 2026 published with Althoff and Reichardt (2026). Appendix A identifies each file exactly and shows that the task ratings reproduce the published occupation scores.

Exposure is a property of the job’s task mix, and two workers in the same occupation receive the same score whatever their skill or their employer. It describes what the technology can do and it is distinct from job loss. A task a model can perform may be automated, may be handed to the model with the worker still doing the rest of the job, or may go on being done by hand where nobody in that workplace has a reason or the means to change. Adoption, electricity, connectivity, product-market competition and the price of labour all sit between an exposure score and any labour-market outcome, and none of them is measured here. Except where Section 6 refers explicitly to the Global Automation Atlas, the words automation exposure or AI exposure in this paper refer to this same score.

The published measure comes in nine versions. Two models, Qwen 2.5 and gpt-oss, are rated under four scenarios each, GPT-4o is rated under one and section 5.4 reports all nine. The headline uses the moderate scenario scored by Qwen 2.5, the version the authors treat as canonical. The choice moves the level of exposure. Four orderings hold across all nine. Agriculture is the least exposed group in every version and far below both non-agricultural groups, clerical support is the most exposed major group in eight of the nine and second to managers in the ninth, formal wage employees score above informal ones in every one, and Nigeria sits above the other two in every one, within the coverage boundary Section 5.2 sets on the distance between Uganda and Tanzania. Inside non-agricultural work, whether wage or own-account sits higher moves with the version, and Section 5.4 reports it.

2.2 Surveys and occupation coding

For each country I use the LSMS-ISA wave that carries coded occupations: the General Household Survey Wave 5 for Nigeria, the National Panel Survey Round 5 for Tanzania, and the National Panel Survey for Uganda. Each records the main-job occupation of working household members at the four-digit ISCO-08 unit group (International Labour Organization, 2012). Ethiopia’s Socioeconomic Panel Survey Wave 5 records only the one-digit major group, too coarse to carry an occupational argument, and it is reported separately in Appendix C.

Coverage of the occupation question differs across the three. In Uganda 7,671 of the 7,688 people with a main-job record carry a four-digit code. Nigeria’s labour module carries a derived work-status variable, and that variable settles who the coded workers are. Of the 7,656 people it records as having worked in the last seven days, for pay, in a non-farm

business or on the farm for market, 7,591 carry a four-digit code and 65 do not. A further 117 coded workers report no work in the last seven days. The sample keeps all 117. Of them, 115 expect to be back in a job within three months, which holds them inside employment on the same definition; the other two are kept for consistency with the survey’s own coding, and dropping them moves no figure in this paper at three decimals. Those two groups are the 7,708 coded Nigerian workers in Table 1. A third group, the 1,839 people whose only work is on the family farm, is asked no occupation question at all, and Section 2.5 sets out how it enters.

For Tanzania, its labour module codes occupation for non-farm jobs only, and farmers enter through the agricultural-labour question. It codes a further 139 workers to the national TASCO classification, for which I found no published correspondence to ISCO-08. And it asks no occupation at all of non-farm business owners without a wage job, which removes 1,789 own-account workers from the Tanzanian sample and means the pooled own-account figures rest on Nigeria and Uganda. Sections 2.5 and 5 take each of these apart.

2.3 From O*NET tasks to survey workers

The chain that carries an exposure score from a task rating to a worker in a Ugandan village runs through four steps, and each loses some information.

*Step 1, tasks to O*NET occupations.* Task ratings are aggregated to each O*NET-SOC 2010 occupation with O*NET task weights, as described above.

*Step 2, O*NET-SOC to SOC-2010.* O*NET-SOC detailed codes roll up to their SOC-2010 base code. This step loses little, given that O*NET-SOC is a refinement of SOC.

Step 3, SOC-2010 to ISCO-08. I use the published ISCO-08 to SOC-2010 correspondence. Mapping is not one-to-one in either direction. Of the 423 ISCO-08 unit groups that receive a score, 127 map to a single O*NET-SOC occupation and 296 map to more than one, with a median of two and a maximum of 35. Where an ISCO code maps to several O*NET occupations I take the unweighted mean of their scores. The unweighted mean treats every O*NET occupation inside an ISCO group as an equally plausible account of what an African worker in that group does. It is a real assumption, and Section 5.3 shows what happens under five alternatives, including one that weights by US employment. Seven codes appearing in the Nigerian data have no direct match. Six take the average of their three-digit parent group and one takes its two-digit parent.

Step 4, ISCO-08 to the survey worker. Workers merge on their four-digit code. The 139 Tanzanian workers coded to TASCO are mapped to ISCO-08 by hand. I found no published correspondence from TASCO to ISCO-08. TASCO adapts ISCO-88, but 18 of the

19 codes involved are TASCO additions with no ISCO-88 counterpart, and the ILO’s ISCO-88 to ISCO-08 table cannot place them. The map was built from the job descriptions the respondents themselves gave, and it is checked in Section 2.5. Of the 24,838 workers, 17,720 (71.3 percent) carry a four-digit code. The other 7,118 are 6,372 family-use own-use farmers, who take a subsistence score described below, 740 market-oriented Tanzanian farmers, who take the major group 6 mean, and six TASCO rows scored above the four-digit level.

2.4 Harmonised variables

For each worker I record age, sex, urban or rural residence, the four-digit occupation and its ISCO major group, and an employment-status category. Agricultural workers are those in ISCO major group 6 or in agricultural elementary occupations, assigned from the occupation code. Everyone else splits into wage employees and own-account workers using the main-job employer question in each survey. In Uganda the question asks about the main job. Appendix B lists the survey variables used for every concept. Employment status, occupation and the formality questions then all describe the same job. The own-account category is a residual. It holds anyone who is not a wage employee, including unpaid family workers and apprentices, and it should be read as own-account and other.

I classify each wage job as formal or informal on a rule the three surveys can all support. A wage job counts as informal when it comes with no written contract and no second marker of formal status, which is pension coverage in Nigeria, income-tax withholding in Tanzania and PAYE deduction in Uganda. This gives a common two-part operational rule, though the second marker differs across surveys. Tanzania and Uganda carry fuller question sets, and Section 4 reports what those give. The formality split covers 4,267 wage workers in three countries.

2.5 Building the estimation sample

Two construction choices decide who counts as employed given that they move the pooled mean more than anything else in the paper. The first is own-use farmers, people who farm only to feed their own household. The 19th International Conference of Labour Statisticians places them outside employment (International Labour Organization, 2013), and the ILOSTAT employment series used in Section 6 follow that boundary. The LSMS-ISA surveys record them and excluding them would drop 6,372 workers, a quarter of the sample, and would reproduce exactly the coverage gap this paper exists to close. I keep them in the headline sample, give them the mean score of the four ISCO subsistence-farming unit groups (6310, 6320, 6330 and 6340), which is 0.019, and report the sample without them

throughout. The boundary turns on the main intended destination of the output. Tanzania’s survey asks it directly, whether the products are intended only for sale, mainly for sale, mainly for family use or only for family use. The 740 who answer only or mainly for sale are market-oriented farmers who also keep some output. They sit inside employment and take the broader ISCO major-6 average of 0.076. Nigeria’s labour module asks the same question in a different form, whether farm output is only for sale, for sale and household use, or only for household use, and if both, what share is sold. Its derived work-status variable puts a person on the family farm only when that farm is their sole work and its output is only or mainly for household use. Mainly means a quarter or less is sold, or half with most kept. Of the 1,839, 985 sell nothing and 854 keep most of the output. For 24 with no farm work in the reference week, the answer comes from the same questions asked about usual farm work. All 1,839 are family-use own-use producers on the same main-destination test applied in Tanzania.

The second is the treatment of the 1,789 Tanzanian non-farm business owners the survey never asks about occupation. These workers cannot be scored at all and are absent from the sample. The consequence is visible in Table 2, where Tanzania’s own-account share is zero. Section 5.2 puts them back under five imputation routes and reports what that does to the country ordering.

Table 1 traces every worker from the household roster to the estimation sample, with the reason for each drop.

Table 1: From the household roster to the estimation sample

Stage	Nigeria	Tanzania	Uganda
Roster respondents	28,256	23,592	16,056
Coded occupation, matched to exposure	7,708	2,208	7,671
TASCO code mapped by hand		139	
Market-oriented farmers		740	
Own-use farmers, family use	1,839	4,533	
Estimation sample	9,547	7,620	7,671
Not recoverable: no occupation question asked		1,789	

Market-oriented farmers, whose output is only or mainly for sale, take the ISCO major-6 average of 0.076, and family-use farmers take the subsistence-group average of 0.019. The final row counts Tanzanian non-farm business owners without a wage job, whom the survey never asks about occupation. The pooled sample is 24,838 workers.

Table 2 reports the sample and its survey-weighted counterpart. The agricultural share runs from 42.2 percent in Nigeria to 76.5 percent in Tanzania, and mean exposure moves against it.

Table 2: Sample composition, unweighted and survey-weighted

Country	<i>N</i>	% female	Mean age	% agriculture		Mean exposure	
				Sample	Weighted	Sample	Weighted
Nigeria 2023	9,547	47.2	38.8	42.2	36.9	0.102	0.107
Tanzania 2020	7,620	44.2	32.4	76.5	80.8	0.050	0.044
Uganda 2019	7,671	49.7	33.4	72.8	64.9	0.055	0.068
All (pooled)	24,838	47.1	35.2	62.2	48.6	0.072	0.090

Mean exposure is the importance-weighted generative-AI score on a 0-1 scale. *N*, % female and mean age are unweighted. Sample columns are unweighted shares of survey rows and are not population estimates. Weights are the cross-sectional household weights supplied with each wave, with complete coverage in all three. Appendix B names them. Agriculture is ISCO major group 6 plus agricultural elementary occupations.

2.6 Weights, survey design and the pooled mean

The paper reports weighted and unweighted figures because they answer different questions. The unweighted mean describes the sample. The weighted mean is a survey-weighted estimate for the analytical population each wave covers, which is the working members of sampled households with a coded occupation. The country estimates in Table 2 carry the population reading. The pooled weighted figure combines three analytical samples taken in three different years, Uganda in 2019, Tanzania in 2020 and Nigeria in 2023, and it leaves out Tanzania’s 1,789 non-farm business owners, whom the survey never asks about occupation.

Country means move little between the two with Nigeria going from 0.102 to 0.107, Tanzania from 0.050 to 0.044 and Uganda from 0.055 to 0.068. Pooling survey rows gives an agricultural share of 62.2 percent and a mean of 0.072. Applying weights and pooling gives 48.6 percent and 0.090. That gap comes from Nigeria, whose population is far larger than Tanzania’s and Uganda’s but whose sample is a similar size. Weighting therefore shifts the pooled figure towards Nigeria’s lower agricultural share and higher exposure. A mean that weights the three countries equally is 0.073.

The unweighted pooled statistics in this paper are described as sample shares. The corresponding population statement for the agricultural share is 48.6 percent across the three countries, and the country-by-country weighted estimates in Table 2 are reported for each of the countries. Exposure is low and graded steeply by occupation under either treatment, and the country ranking is the same: Nigeria, then Uganda, then Tanzania.

All three surveys are stratified multi-stage samples, and the intervals reported below account for stratification and clustering by primary sampling unit (PSU). Intervals around unweighted sample means do not turn those means into population estimates. The term population estimate is kept for the weighted figures. Nigeria and Tanzania supply cluster and

stratum codes directly. Uganda supplies neither, and the parish stands in for the enumeration area, which is the identifier any household file carries. Appendix E gives the design per country and shows what the substitution costs. Standard errors that allow for strata and clusters run from 0.96 to 1.55 times their household-clustered counterparts, and the ratio is largest for groups whose members share villages.

3 How much exposure, and where

The pooled sample mean is 0.072, with a 95 percent interval of 0.069 to 0.074. The weighted population estimate is 0.090. Either way the average worker in these three countries holds a job in which well under a tenth of the task importance sits in tasks a language model is rated able to perform. More than half of all workers, 57.5 percent, score below 0.05, which is close to nothing, and 68.5 percent work in agricultural or elementary occupations. The most exposed tenth of workers holds 43 percent of the sum of worker exposure scores, unweighted.

Table 3 reports worker shares and mean exposure by ISCO major group. The gradient is steep and it runs along familiar white-collar lines. Clerical support work sits above everything else as its mean of 0.615 is nearly double the next group, technicians at 0.363, and about twenty times the score for skilled agriculture. This fits what is already known about large language models as they are at their strongest on text, records, correspondence, scheduling and routine information handling, which is close to a full description of clerical work. Eloundou et al. (2024) find the same peak in US data.

Clerical support holds 0.6 percent of the pooled sample. Technicians add 1.6 percent. The two are the most exposed groups and account for 2.2 percent of the workers. Skilled agriculture, at 0.031, holds 57.8 percent of the sample, and elementary occupations, the least exposed group at 0.028, hold another 10.7 percent. Figure 1 sets the two columns side by side. Every group with a long exposure bar has a short worker bar, and the one group that fills the worker panel sits at the bottom of the exposure panel. The occupations generative AI reaches furthest into are the occupations these economies barely have and that is the mechanism behind the low average.

4 Who is exposed

Table 4 splits the sample by work status. Agriculture sits at 0.029. Non-agricultural own-account workers average 0.134 and non-agricultural wage employees 0.150. Estimated directly, the wage premium over own-account work is 0.016 with a standard error of 0.004, and 0.033 with a standard error of 0.005 once country fixed effects absorb the different country

Table 3: Worker share and mean generative-AI exposure by ISCO major group

ISCO major group	Share of workers	Mean exposure	95% interval
Clerical support	0.6%	0.615	[0.590, 0.639]
Technicians and associates	1.6%	0.363	[0.339, 0.388]
Managers	4.8%	0.209	[0.204, 0.215]
Professionals	4.6%	0.161	[0.151, 0.170]
Plant and machine operators	3.4%	0.144	[0.138, 0.149]
Service and sales	11.5%	0.142	[0.138, 0.147]
Craft and trades	5.0%	0.051	[0.046, 0.056]
Skilled agriculture	57.8%	0.031	[0.030, 0.032]
Elementary occupations	10.7%	0.028	[0.026, 0.030]

Rows are sorted by mean exposure. Shares are of all 24,838 pooled workers, including 7,112 farmers without a directly coded four-digit occupation, all in major group 6. Of these, 6,372 are family-use own-use producers and 740 are market-oriented Tanzanian farmers. Intervals are 95 percent and account for stratification and PSU clustering. The means are unweighted, and the intervals do not make them population estimates.

mixes. Both are far from zero and both are small next to the gap of about 0.11 that separates either group from agriculture. On the weighted basis the ordering is the same.

Table 4: Worker share and mean generative-AI exposure by work status

Work status	Share of workers	Mean exposure	95% interval	Weighted
Agriculture	62.2%	0.029	[0.028, 0.031]	0.037
Own-account, non-agriculture	20.6%	0.134	[0.130, 0.138]	0.128
Wage employee, non-agriculture	17.2%	0.150	[0.144, 0.156]	0.169

Agriculture is assigned from the occupation code. The remaining workers split on the survey's main-job employer question, with own-account as the residual category. Tanzania contributes no own-account workers, as its survey asks no occupation of non-farm business owners, so that row rests on Nigeria and Uganda. Intervals are 95 percent and account for the survey design.

The surveys put the formality questions to wage employees only, which leaves the share of own-account workers who are informal unmeasured. Under the ILO definition an own-account worker counts as informal when the enterprise does, and status on its own settles nothing (International Labour Organization, 2023). What the data do say is that non-agricultural own-account workers score 0.134, close to wage employees and well above farmers. The measured distance between a subsistence farmer and everyone else comes from the task content the measure assigns to farming, and the presence of a contract plays no part in it.

The formality question sharpens this. Within non-agricultural wage employment, workers with no written contract and no second marker of formal status average 0.104, while formal wage workers average 0.213. Estimated directly on wage workers with country fixed effects,

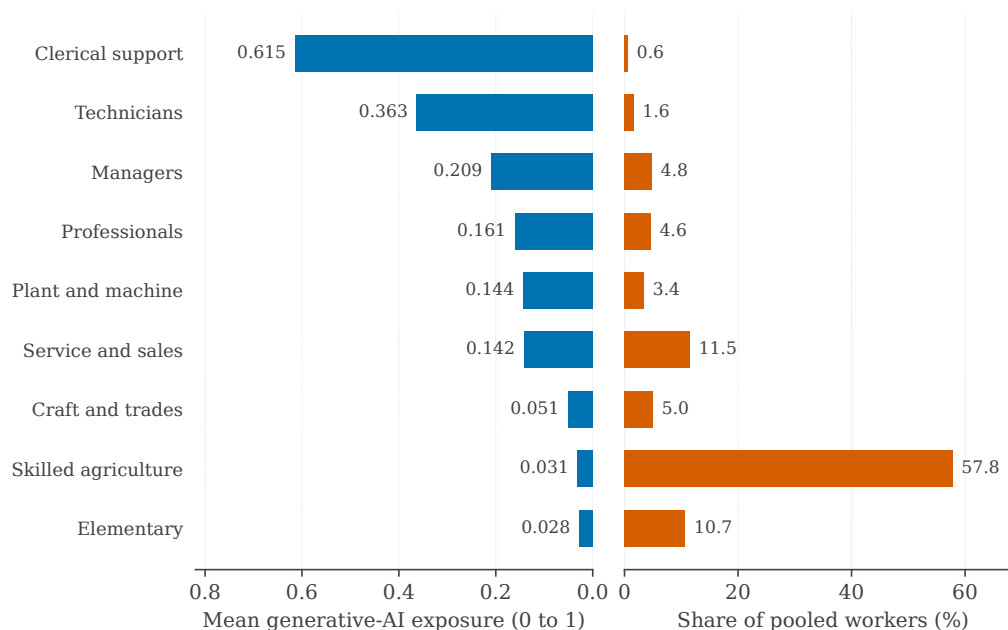


Figure 1: Mean generative-AI exposure and share of workers by ISCO major group, pooled sample, ordered by exposure. The most exposed major groups hold the smallest shares of workers.

the informal-formal contrast is -0.102 with a standard error of 0.006 . Formal wage workers are about twice as exposed as informal ones, and the pattern holds in all three countries, with 0.234 against 0.121 in Nigeria, 0.196 against 0.097 in Tanzania and 0.194 against 0.102 in Uganda. Formality therefore carries a strong signal inside wage work. Measured informality among wage workers is 40.4 percent in Nigeria, 67.3 percent in Tanzania and 65.4 percent in Uganda. Where a country's survey supports a fuller rule the level shifts and the exposure gap holds. Tanzania reads 81.3 percent informal on the tax question alone, and Uganda 64.7 percent on contract, pension and PAYE together.

Two divides therefore run through these labour markets, and per worker they are much the same size. Agriculture sits 0.121 below non-agricultural wage employment. Formal wage work sits 0.109 above informal wage work. What separates the two divides is reach as the agricultural one sorts $15,441$ workers, 62.2 percent of the sample, onto the low side. The formality one operates inside non-agricultural wage employment, 17.2 percent of the sample, and only $1,801$ workers sit on its formal side. Agriculture is what holds the aggregate down, while formality is a steep gradient across a thin slice. Treating agriculture, own-account work and informality as one undifferentiated low-exposure mass would miss the second divide entirely.

Urban workers average 0.120 against 0.055 for rural workers, a gap of 0.064 that is wide in

every country and that survives country fixed effects at 0.062. Men average 0.077 and women 0.066. The sex difference is estimated and small. At 0.011 it is a sixth of the urban-rural gap and a tenth of the agricultural gap.

Table 5 puts these differences in one place and asks how much of each is composition across countries. Column 1 is the raw status comparison. Column 2 adds country fixed effects, and column 3 adds residence, sex and age. The status gaps narrow a little and survive. Relative to agriculture, own-account work carries 0.088 more exposure and wage employment 0.111 more, with country, residence, sex and age held fixed. Figure 2 shows every group mean with its interval.

Table 5: Generative-AI exposure and work status

	(1)	(2)	(3)
Own-account, non-agriculture	0.105*** (0.002)	0.093*** (0.003)	0.088*** (0.003)
Wage employee, non-agriculture	0.121*** (0.003)	0.120*** (0.003)	0.111*** (0.003)
Urban			0.015*** (0.002)
Female			-0.005*** (0.001)
Age			0.0003*** (0.00005)
Country fixed effects		✓	✓
Observations	24,838	24,838	24,838
R^2	0.270	0.277	0.285

Ordinary least squares. The dependent variable is the worker’s generative-AI exposure score. Agriculture is the omitted status. Regressions are unweighted and describe conditional relationships in the analytical sample. Weighted by the survey expansion weights, column (2) gives 0.080 (0.003) for own-account and 0.129 (0.006) for wage work. Standard errors in parentheses account for the stratified multi-stage survey design. In the matching regression run on wage workers alone, informal status carries -0.102 (0.006) with country fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5 Robustness

Three features of the construction could carry the results on their own: which workers count as employed, the unweighted mean used when an ISCO code maps to several O*NET occupations, and the choice among the nine published versions of the exposure measure. This section tests each.



Figure 2: Mean generative-AI exposure by group, with 95 percent intervals accounting for the survey design. Most intervals are narrower than the marker. The dashed line is the pooled mean.

5.1 The sample definition

Table 6 recomputes the headline figures on four samples. The strict column keeps the workers whose occupation came from the survey’s own coding, 17,726 in all. Of these, 17,720 carry a four-digit ISCO-08 code; six are Tanzanian TASCO rows mapped to a sub-major group, which scores them without giving them a unit code.

Leaving out the family-use own-use farmers, which is what the ILO employment boundary implies, raises the pooled mean from 0.072 to 0.090 and cuts the agricultural share from 62.2 to 49.1 percent. Keeping only coded occupations gives almost the same figures. Dropping the 139 hand-mapped TASCO rows moves the pooled mean by 0.001. The occupational gradient, which is the paper’s result, is untouched in every column, where the two non-agricultural status means are identical to three decimals throughout, as none of the affected workers is in them.

The score given to the own-use farmers deserves a test of its own, as it sets the exposure of a quarter of the sample. In the headline the 6,372 family-use farmers receive 0.019, the average of the four ISCO subsistence unit groups. Table 7 recomputes the pooled mean under four alternatives, from zero to the highest-scoring major-6 occupation. The pooled level does depend on this choice and the ranking of groups does not. At every value, agriculture stays far below both non-agricultural groups, and the full range of defensible pooled means, 0.067

Table 6: Robustness to the sample definition

	Headline	Family-use farmers out	Coded occupations only	TASCO rows left out
Workers	24,838	18,466	17,726	24,699
Mean exposure	0.072	0.090	0.090	0.071
Mean exposure, weighted	0.090	0.110	0.111	0.090
% agriculture	62.2	49.1	47.0	62.5
% scoring below 0.05	57.5	42.8	44.6	57.8
Agriculture, mean	0.029	0.037	0.033	0.029
Own-account non-agric., mean	0.134	0.134	0.134	0.134
Wage non-agric., mean	0.150	0.150	0.150	0.151

The second column applies the ILO employment boundary (International Labour Organization, 2013) by dropping the 6,372 family-use own-use farmers and keeping Tanzania’s 740 market-oriented farmers. The third keeps only workers with a directly coded four-digit occupation. The fourth drops the 139 Tanzanian workers whose TASCO codes were mapped to ISCO-08 by hand.

to 0.093, leaves the pooled mean below a tenth throughout.

Table 7: Pooled exposure under alternative scores for the own-use farmers

Score	Source	Pooled mean	Weighted	Agriculture mean
0.000	Lowest major-6 code	0.067	0.086	0.022
0.019	Subsistence groups (headline)	0.072	0.090	0.029
0.076	Major-6 average	0.086	0.103	0.053
0.102	Highest major-6 code	0.093	0.109	0.063

The score is applied to the 6,372 family-use own-use farmers in Nigeria and Tanzania. Changing the score of Tanzania’s 740 market-oriented farmers moves the pooled mean by about 0.002 at most.

5.2 Tanzania’s uncounted business owners

Tanzania’s survey asks no occupation of 1,789 non-farm business owners, who therefore carry no score and sit outside the estimation sample, which is why Tanzania’s own-account share in Table 2 is zero. The missing workers all belong to the country at the bottom of the ranking, and none of the tests above can see them.

Table 8 puts them back. Each of the 1,789 carries its own household expansion weight, joined from the file the rest of the Tanzanian sample already uses, and averaging 2,628 against a Tanzanian mean of 2,736. Four flat scores bound the answer: zero, the agricultural mean of 0.029, the own-account non-agricultural mean of 0.128 and the wage non-agricultural mean of 0.169. A fifth route scores them by industry. Of the 1,789, 1,555 (87.1 percent) link to a household enterprise carrying an ISIC section code, and each takes the weighted mean

exposure of Ugandan non-agricultural own-account workers in the same section. That route implies 0.162, inside the flat range. These workers stay out of the headline sample, since any score given to them is an assumption.

The pooled mean barely moves, from 0.087 at a score of zero to 0.093 at the highest imputation, against a headline of 0.090. Tanzania stays below Uganda at every score, while the margin closes. At zero it is 0.033. At the wage non-agricultural mean it is 0.001, and the two countries' 95 percent design-based intervals overlap. The industry route does the same. The country order Nigeria, Uganda, Tanzania therefore holds as a ranking of point estimates under every crosswalk rule, in all nine rating versions and against this add-back. The distance between Uganda and Tanzania is smaller than the coverage gap in one of them, and what this paper takes from the cross-country comparison is Nigeria's distance from the other two.

Table 8: Adding Tanzania's 1,789 uncounted business owners back

Score	Source	Pooled, weighted	Tanzania, weighted	Tanzania, 95% interval	Margin over Uganda
0.000	Zero	0.087	0.036	0.033–0.038	0.033
0.029	Agriculture mean	0.088	0.041	0.039–0.043	0.027
0.128	Own-account non-agric.	0.092	0.059	0.056–0.062	0.009
0.169	Wage non-agric.	0.093	0.067	0.063–0.070	0.001
0.162	Industry match	0.093	0.065	0.062–0.068	0.003

Each of the 1,789 carries its own household expansion weight. Uganda's weighted mean is 0.068, with a 95 percent interval of 0.063 to 0.073, and is the same in every row. Intervals are design-based, with strata and primary sampling units rebuilt from the Tanzanian household roster. The margin is Uganda's weighted mean less Tanzania's. In the last two rows the two intervals overlap. Giving the 1,789 the mean Tanzanian weight instead of their own moves every figure in the table by 0.001 at most.

5.3 The crosswalk aggregation

The unweighted mean over O*NET occupations within an ISCO group is the other consequential choice. Table 9 recomputes the headline figures under five alternatives: the median, the minimum and the maximum of the linked scores, a mean weighted by US employment, and 2,000 draws that pick one linked occupation per ISCO code at random.

The employment weights come from the BLS Occupational Employment and Wage Statistics for May 2018 (U.S. Bureau of Labor Statistics, 2019), the last release coded to the 2010 SOC and therefore the one that matches the crosswalk. They reach 422 of the 423 ISCO codes. This alternative answers a fair objection to the unweighted mean, which is that O*NET subdivides some SOC occupations into several detailed codes and leaves others

whole, letting an arbitrary filing decision move the average. It also imports the US occupational size distribution into an African measure, which is why it appears here as an alternative.

Individual clerical ISCO codes run from 0.149 to 0.858, against a group mean of 0.615, the group average is taken over codes that differ widely. The alternatives in Table 9 nonetheless leave the clerical mean between 0.543 and 0.684, a narrower proportional range than agriculture’s. The size of the clerical peak should be treated as approximate and its position at the top of the distribution is somewhat secure.

Clerical support stays between 0.54 and 0.68, and at least thirteen times the skilled-agriculture score. The country order is Nigeria, Uganda, Tanzania, and the draws’ 95 percent ranges for the three countries do not overlap. Agriculture is the least exposed group. The separation of Uganda from Tanzania carries a boundary these draws cannot see, and Section 5.2 states it.

The formality gap inside wage work holds under every rule as well. It stays between 0.094 and 0.120 at the extreme bounds, and formal employees score above informal ones in every one of the 2,000 draws. The level shifts by about a third either way at the bounds, and across the draws the pooled weighted mean has a standard deviation of 0.005, more than twice its sampling standard error. And the wage premium over own-account work, already the smallest gap in the paper, falls to zero at the minimum, though wage workers score higher in 99.3 percent of draws and the employment-weighted rule widens the gap to 0.020. That contrast is the least secure result reported here, and Section 5.4 gives a second reason to treat it with care.

5.4 The exposure measure

Althoff and Reichardt (2026) publish nine versions of the task ratings. Two models, Qwen 2.5 and gpt-oss, are rated under four scenarios each, and GPT-4o is rated under one. Table 10 rescores every worker under each.

The structure survives all nine. The country order is Nigeria, Uganda, Tanzania in every one, weighted and unweighted, within the boundary Section 5.2 sets on the distance between Uganda and Tanzania. Agriculture is the least exposed employment group in every one. Clerical support is the most exposed major group in eight of the nine, and in the ninth it is second to managers. Formal wage employees score above informal ones in every one, by 0.067 to 0.155. Rank correlations with the headline run from 0.80 to 0.96 across the 430 ISCO codes.

The pooled weighted mean runs from 0.045 to 0.489 across the nine. The model matters

Table 9: Sensitivity to the ISCO-to-O*NET aggregation rule

	Mean (headline)	Emp. weighted	Median	Min.	Max.	Draws, 95% range
Pooled, weighted	0.090	0.080	0.089	0.058	0.126	0.081–0.100
Nigeria, weighted	0.107	0.095	0.106	0.070	0.147	0.095–0.118
Uganda, weighted	0.068	0.057	0.066	0.042	0.098	0.058–0.078
Tanzania, weighted	0.044	0.041	0.041	0.024	0.068	0.036–0.053
Agriculture	0.029	0.020	0.029	0.008	0.050	0.018–0.041
Own-account non-agric.	0.134	0.126	0.130	0.105	0.169	0.121–0.147
Wage non-agric.	0.150	0.146	0.146	0.106	0.206	0.138–0.164
Wage minus own-account	0.016	0.020	0.016	0.000	0.036	0.003–0.029
Clerical support	0.615	0.613	0.617	0.543	0.684	0.568–0.661
Skilled agriculture	0.031	0.021	0.031	0.008	0.053	0.019–0.044
Wage, formal	0.213	0.206	0.211	0.160	0.275	0.198–0.229
Wage, informal	0.104	0.102	0.098	0.066	0.155	0.090–0.120
Formal minus informal	0.109	0.104	0.113	0.094	0.120	0.096–0.124

Minimum and maximum are extreme bounds in which every ISCO code takes its lowest or highest linked score at once. Group means are unweighted unless marked. Employment weights are US national employment from U.S. Bureau of Labor Statistics (2019), applied at the 6-digit SOC level and split evenly among the O*NET codes in each base; they reach 422 of the 423 ISCO codes, and the remaining code keeps the simple mean. The draws pick one linked O*NET occupation per ISCO code at random, 2,000 times. The last three rows rest on the 4,267 wage workers whose formality status is observed, a smaller base than the rows above them.

as much as the scenario. Under every scenario gpt-oss gives a higher pooled mean than Qwen. The share of workers scoring below 0.05 runs from 0.1 percent to 71.6 percent. The level of exposure, and about how many workers sit near zero reported in this study, belongs to the moderate Qwen scenario. The occupational gradient and the gap between agriculture and both non-agricultural groups hold across all nine, and neither depends on it.

Wage workers score above own-account workers under three of the nine versions, and own-account workers score higher under the other six, by up to 0.114. With the crosswalk bounds in Section 5.3, we conclude that non-agricultural wage and own-account work are close together and far above agriculture.

6 From within-country to cross-country

The within-country evidence document that exposure is low because of what workers do, but how much of the low exposure the occupational mix has on its own outside the three countries. In Table 11, the exposure score attached to each ISCO major group is held at the value these three surveys imply, and the employment shares are swapped for a high-income country's. The shares come from ILOSTAT for the United Kingdom, Germany and France, each of which reports ISCO-08 with under 2 percent of employment unclassified. In the

Table 10: Exposure under the nine published versions of the task ratings

Model	Scenario	Rank corr.	Pooled, weighted	Agri- culture	Own- account	Wage	Clerical	Formal less informal
Qwen 2.5	Moderate (headline)	1.00	0.090	0.029	0.134	0.150	0.615	0.109
gpt-oss	Moderate	0.84	0.256	0.215	0.283	0.275	0.686	0.074
Qwen 2.5	Slow	0.94	0.045	0.011	0.066	0.085	0.409	0.081
gpt-oss	Slow	0.88	0.195	0.149	0.243	0.219	0.650	0.094
Qwen 2.5	Rapid	0.96	0.200	0.082	0.325	0.263	0.784	0.155
gpt-oss	Rapid	0.83	0.489	0.384	0.602	0.488	0.790	0.143
Qwen 2.5	No scenario	0.96	0.061	0.013	0.094	0.117	0.473	0.105
gpt-oss	No scenario	0.80	0.103	0.057	0.143	0.124	0.474	0.067
GPT-4o	No scenario	0.87	0.117	0.058	0.167	0.148	0.551	0.086

Rank correlation is with the headline scores across the 430 ISCO-08 codes. Group means are un-weighted. Every version is rescored through the same task-to-occupation and occupation-to-ISCO rules as the headline. The scenario sets how fast the rater assumes model capability advances. The last column is the formal wage mean less the informal wage mean, and rests on the 4,267 wage workers whose formality status is observed, a smaller base than the other columns.

third column, its score vector drops the family-use own-use farmers, so the scores and the ILOSTAT shares describe the same employed population. On this construction the three economies studied in the paper average 0.088. The United Kingdom’s mix raises that to 0.209, Germany’s to 0.227 and France’s to 0.203. The average African country moves from 0.104 to 0.209 on the United Kingdom’s mix. The full-sample survey scores give 0.080 to 0.208, the crosswalk scores give the same pattern at a higher level, and the three benchmarks agree.

The full-sample baseline of 0.080 differs from the headline weighted mean of 0.090, and the whole of the difference is the source of the employment shares. The table needs a share for all nine major groups in every country, including the benchmarks. Only ILOSTAT covers all of them, and the table uses it throughout. Aggregation adds nothing to the gap. Each group’s score is itself the weighted mean of the worker scores inside it. Applying the survey’s own major-group shares to the same score vector returns 0.090 exactly, to the tenth decimal. What moves the number is that ILOSTAT and the surveys place workers in different groups. ILOSTAT puts 0.7 percent of these three countries’ employment in major group 1 against 8.2 percent in this sample, which on its own accounts for -0.015 of the -0.011 difference, and service and sales runs the other way, 23.3 percent against 15.8 percent, for $+0.009$. Every other group contributes under 0.004. Major group 1 in this sample is 1,202 workers, 89 percent of them own-account and non-agricultural by weight, and 72 percent of that weight sits in one unit code, retail and wholesale trade managers. The gap is about where a shopkeeper is coded.

Group by group, the rise sits at the top of the occupational distribution. For the average

African country, managers, professionals, technicians and clerical support together contribute 0.133 to the rise from 0.104 to 0.209, and the smaller share in skilled agriculture takes 0.021 of it back. The gain comes from the jobs a high-income workforce holds. Agriculture’s retreat works the other way, since those workers carried a score close to zero to begin with.

Changing nothing but where people work more than doubles exposure and that is the size of the composition result. The exercise holds the occupation scores fixed, such that it says what the mix does and leaves open whether the same occupation has the same tasks in a rich country. It also gives no share of the gap between these economies and a high-income one, since the benchmark’s exposure under its own occupation scores is not estimated. And the counterfactual level, about a fifth, still sits far below the clerical figure of 0.615, so a high-income mix moves these economies a long way up a distribution whose top most workers never reach.

Table 11: Exposure under a high-income occupational mix

Occupational mix	Crosswalk mean scores	Survey scores	Survey scores, own-use out
These three economies, own mix	0.113	0.080	0.088
Average African country, own mix	0.133	0.098	0.104
United Kingdom	0.261	0.208	0.209
Germany	0.256	0.227	0.227
France	0.242	0.203	0.203

Each cell is the sum over ISCO major groups 1 to 9 of an employment share times a group exposure score. The score vector is held fixed down each column and the employment shares change down the rows. Crosswalk mean scores average the O*NET-linked score over the ISCO codes in each major group. Survey scores are the weight-averaged worker scores estimated here. The third column, the headline, drops the family-use own-use farmers from the score vector, matching the ILO employment boundary the ILOSTAT shares follow. Employment shares come from International Labour Organization (2026), taking each country’s latest year from 2015 on, which is 2025 for all three benchmarks and 2024, 2024 and 2021 for Nigeria, Tanzania and Uganda against survey years of 2023, 2020 and 2019. The first row weights the three countries by their survey expansion weights, and the second is the simple average across African countries. The 0.080 in the second column rests on ILOSTAT employment shares, which is why it differs from the survey-weighted headline of 0.090.

The counterfactual rests on one of two country-level measures that reach most of the continent. The two disagree in a way that is worth setting out, and the rest of this section does that. The first is the Global Automation Atlas (Garg et al., 2026), which reports the share of tasks rated exposed for 32 African economies and it carries no separate generative-AI channel. I rebuild three shares from the underlying task-country file: tasks exposed through AI, tasks exposed through learned content transformation, which is the closest available match to generative models, and tasks exposed with no AI involved. AI accounts for 46 percent of exposed tasks in the average African country.

The second measure takes employment by ISCO major group from ILOSTAT (International Labour Organization, 2026) for 45 African countries and applies the worker-level exposure scores estimated in this paper. A country’s figure then moves only with its occupational mix. The Atlas figure moves for a different reason, as its classifier judges the same task differently across countries while employment never enters. The two are related and far from identical, correlating at 0.59 to 0.69 across the 28 countries both cover.

Table 12 sets the worker-level mean for each of the three countries against both. Uganda matches almost exactly, at 0.068 in the survey and 0.067 in the ILOSTAT rebuild. Tanzania is sensitive to the employment boundary. ILOSTAT puts 52 percent of Tanzanian workers in agriculture and the survey 74 percent, as ILOSTAT follows the ILO boundary and leaves out people who farm only for their own use. Excluding own-use producers raises the survey estimate from 0.044 to 0.084, above the ILOSTAT figure of 0.066 on the same boundary. Differences in occupational composition between the survey and ILOSTAT remain after the boundary is harmonised. Tanzania is evidence about sensitivity to definitions.

Table 12: Worker-level and country-level generative-AI exposure

Country	Survey, weighted		ILOSTAT rebuild	
	Headline	Own-use out	Survey scores	Own-use out
Nigeria 2023	0.107	0.125	0.088	0.095
Uganda 2019	0.068	0.068	0.067	0.079
Tanzania 2020	0.044	0.084	0.057	0.066

The ILOSTAT rebuild applies the survey-based major-group scores to each country’s employment shares from International Labour Organization (2026). Own-use out applies the ILO employment boundary (International Labour Organization, 2013) on both sides. Nigeria’s ILOSTAT series leaves 23 percent of workers unclassified.

Across the wider cross-section the structural pattern is reported in Figure 3. Country exposure falls with the informal employment rate and with the agricultural self-employment share, and rises with income per head. Figure 3 plots exposure against informality and marks the three countries observed at the worker level. The two measures describe this cross-section differently, and Table 13 shows how. Informality predicts lower exposure under both. Conditioning on agriculture and income, the Atlas measure keeps a large informality coefficient of -0.25 , while under the ILOSTAT rebuild informality falls to about -0.03 and cannot be told apart from zero, with the agricultural share carrying the link.

Informality and agricultural employment are hard to separate across 27 country averages, and which of the three predicts exposure best depends on the measure. On the same 27 countries, put on a common scale, the agricultural share is the strongest of the three under the ILOSTAT rebuild, with a standardised coefficient of -0.49 against -0.18 for informality

and 0.30 for income. Under the Atlas measure informality and income are close and neither leads on both metrics. Table 16 reports the standardised coefficients and the incremental R-squared of each variable. The occupation-based measure, which is the one built from the worker-level evidence in this paper, puts the weight on where employment sits. Part of that is built in, as a score made from occupation shares must move with the share in major group 6, which is why Table 11 states the size of the composition effect directly. These are 27 country averages describing a cross-section, and no coefficient here measures what moving one of the variables would do. The Atlas informality result is concentrated in the AI-material component of its index. For tasks exposed without AI the same coefficient is -0.006 , which is small. That is a fact about how the Atlas classifier judges countries, and the cross-country evidence should not be presented as showing that informality insulates workers beyond what their occupations already imply.

Table 13: Country-level AI exposure and employment structure

	Atlas, exposed through AI		ILOSTAT rebuild	
	Alone	Full model	Alone	Full model
Informal employment rate	-0.482 (0.046)	-0.242 (0.049)	-0.130 (0.024)	-0.035 (0.020)
Agricultural self-employment		-0.093 (0.048)		-0.054 (0.020)
Log GDP per head		0.045 (0.011)		0.009 (0.005)
Countries	28	28	27	27

Ordinary least squares across African economies. The Atlas outcome is the share of tasks rated exposed through AI, rebuilt from the task-country file of Garg et al. (2026). It counts a task as exposed when AI is materially involved in the automation route, which is broader than generative AI. The narrower generative-AI route appears in Appendix D. The ILOSTAT outcome applies the survey-based major-group scores to each country’s employment shares. Informal employment and agricultural self-employment enter as shares of employment on a 0-1 scale. Heteroskedasticity-robust standard errors in parentheses. The full model holds all three variables at once. Appendix D reports the other Atlas outcomes and the alternative score vectors.

7 Limitations

The exposure measure is built outside Africa. It scores the task composition of US occupations, and it reaches an African worker only through a crosswalk. A Ugandan shopkeeper and a US retail salesperson share an ISCO code and receive the same score, though the two jobs differ in tools, scale and what the worker actually spends the day doing. If African

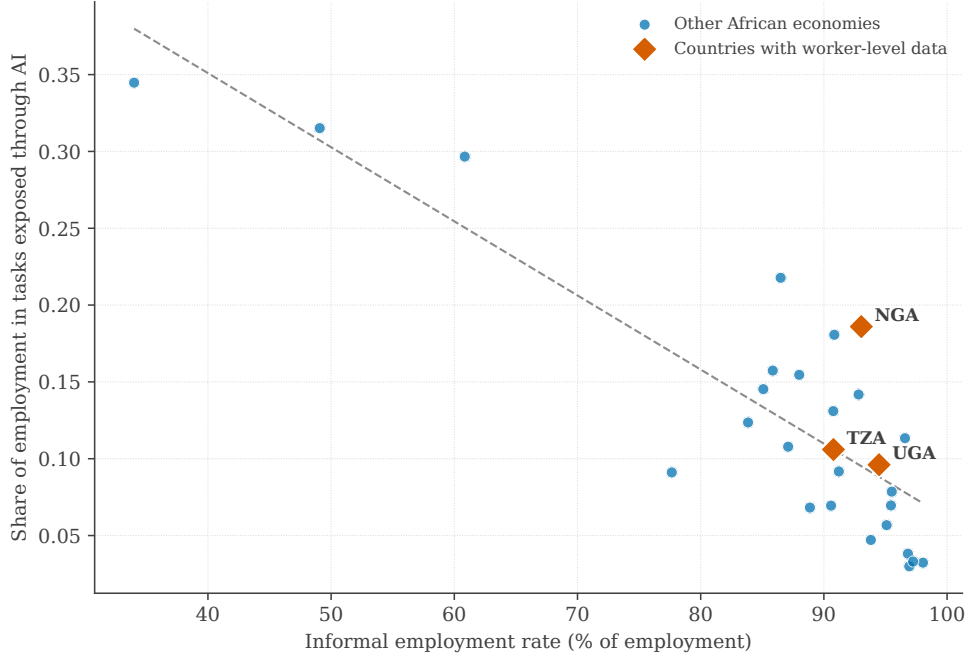


Figure 3: Share of tasks rated exposed through AI against the informal employment rate, 28 African economies. The dashed line is a linear fit.

jobs inside a given occupation are more manual than their US counterparts, the measure overstates exposure here, and the direction of that bias runs against the paper’s finding.

The level of exposure is a property of the version of the measure used. Across the nine published versions the pooled weighted mean runs from 0.045 to 0.489. Section 5.4 reports all nine, and every level statement in the paper is tied to the moderate Qwen scenario.

The occupation coding is uneven. Tanzania codes occupation for non-farm work only, and its farmers enter through the own-use route. It also asks no occupation of non-farm business owners without a wage job, which removes 1,789 own-account workers and leaves the pooled own-account figures resting on Nigeria and Uganda. Nigeria leaves 65 of the 7,656 people its labour module records as working without an occupation code. Ethiopia offers only one-digit codes and is reported separately.

Whether non-agricultural wage or own-account work is more exposed is not settled here. The gap is 0.016, it falls to zero under the extreme crosswalk bound, and it reverses under six of the nine versions of the exposure measure.

Uganda’s survey supplies no enumeration-area identifier, and the parish stands in as the primary sampling unit. For 751 of its 3,094 households, tracked movers and split-offs, and the resulting standard errors understate clustering. Appendix E gives district-level clustering as a bound, which moves the standard error of Uganda’s mean from 0.0017 to 0.0025.

The design is descriptive throughout. We do not identify a causal effect of occupational structure on exposure, and the cross-country regressions are associations among 27 country averages, several of which are close to being the same variable measured differently.

8 Conclusion

This paper measures generative-AI task exposure across three Sub-Saharan African economies with household survey data that distinguish agricultural, own-account and wage work, record wage-job formality, and also include own-use producers outside the conventional employment boundary. Exposure is low, spread very unevenly across occupations, and concentrated in a small set of white-collar jobs, on the moderate 2030 ratings scored by Qwen 2.5. Agricultural workers, who are 62.2 percent of the pooled sample and 48.6 percent of the weighted population, average 0.029 against 0.150 for non-agricultural wage employees. Clerical support, the most exposed major occupational group by a wide margin at 0.615, employs 0.6 percent of the sample. Across African economies, exposure is negatively associated with agricultural employment. The strength and interpretation of that relationship depend on how country exposure is measured.

Formality tracks exposure inside wage work, where formal employees score about twice what informal ones do. Own-account formality is unmeasured. The measured distance between farming and everything else comes from the task content the measure assigns to farming.

The level of exposure is a property of the measure. Across the nine published versions of the task ratings the pooled figure moves by a factor of ten. Agriculture is the least exposed group in every version, and far below both non-agricultural groups. Clerical support is the most exposed major group in eight of the nine and second to managers in the ninth. Formal wage employees score above informal ones in every one. Nigeria sits above the other two in every one. Whether wage or own-account work sits higher inside non-agricultural employment moves with the version. And the distance between Uganda and Tanzania is smaller than Tanzania's own coverage gap, so the country ranking carries Nigeria's distance from the other two and no more.

And exposure is distinct from displacement. Under the moderate 2030 scenario, the occupational reach of generative AI in these economies falls disproportionately on a small share of workers, which says where the technology could bite first. Whether firms will adopt it, whether output per worker will rise, and whether anyone will lose a job are separate questions, and answering them needs data on adoption and outcomes that these surveys do not have.

Declaration of generative AI in the manuscript preparation process

Statement: During the preparation of this work the author used Claude Opus 5 for code generation and debugging and for prose editing. The author designed the analysis, independently audited the code, and reproduced all outputs. The author takes full responsibility for the content of the manuscript.

References

- Daron Acemoglu and Pascual Restrepo. Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30, 2019.
- Daron Acemoglu and Pascual Restrepo. Tasks, automation, and the rise in u.s. wage inequality. *Econometrica*, 90(5):1973–2016, 2022.
- Cristian Alonso, Andrew Berg, Siddharth Kothari, Chris Papageorgiou, and Sidra Rehman. Will the ai revolution cause a great divergence? IMF Working Paper WP/20/184, International Monetary Fund, 2020.
- Lukas Althoff and Hugo Reichardt. Task-specific technical change and comparative advantage. NBER Working Paper 35353, National Bureau of Economic Research, 2026.
- David H. Autor, Frank Levy, and Richard J. Murnane. The skill content of recent technological change: An empirical exploration. *Quarterly Journal of Economics*, 118(4):1279–1333, 2003.
- Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond. Generative ai at work. NBER Working Paper 31161, National Bureau of Economic Research, 2023.
- Mauro Cazzaniga, Florence Jaumotte, Longji Li, Giovanni Melina, Augustus J. Panton, Carlo Pizzinelli, Emma Rockall, and Marina Mendes Tavares. Gen-ai: Artificial intelligence and the future of work. Staff Discussion Note SDN/2024/001, International Monetary Fund, 2024.
- Gabriel Demombynes, Jörg Langbein, and Michael Weber. The exposure of workers to artificial intelligence in low- and middle-income countries. Policy Research Working Paper 11057, World Bank, Washington, DC, 2025.

- Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. Gpts are gpts: Labor market impact potential of large language models. *Science*, 384(6702):1306–1308, 2024.
- Edward Felten, Manav Raj, and Robert Seamans. Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12):2195–2217, 2021.
- Carl Benedikt Frey and Michael A. Osborne. The future of employment: How susceptible are jobs to computerisation? *Technological Forecasting and Social Change*, 114:254–280, 2017.
- Prashant Garg, Tommaso Crosta, and Jasmin Baier. Global automation atlas. arXiv preprint arXiv:2605.17086, 2026.
- Paweł Gmyrek, Janine Berg, and David Bescond. Generative ai and jobs: A global analysis of potential effects on job quantity and quality. ILO Working Paper 96, International Labour Organization, 2023.
- Paweł Gmyrek, Hernan Winkler, and Santiago Garganta. Buffer or bottleneck? employment exposure to generative AI and the digital divide in Latin America. ILO Working Paper 121, International Labour Office and World Bank, Geneva, 2024.
- Paweł Gmyrek, Janine Berg, Karol Kamiński, Filip Konopczyński, Ania Ładna, Balint Nafradi, Konrad Rosłaniec, and Marek Troszyński. Generative ai and jobs: A refined global index of occupational exposure. ILO Working Paper 140, International Labour Organization, Geneva, 2025.
- Douglas Gollin, David Lagakos, and Michael E. Waugh. The agricultural productivity gap. *Quarterly Journal of Economics*, 129(2):939–993, 2014.
- Xiang Hui, Oren Reshef, and Luofeng Zhou. The short-term effects of generative artificial intelligence on employment: Evidence from an online labor market. Working Paper 10601, CESifo, 2023.
- International Labour Organization. International standard classification of occupations: Isco-08. ILO, Geneva, 2012.
- International Labour Organization. Resolution concerning statistics of work, employment and labour underutilization. 19th International Conference of Labour Statisticians, Geneva, 2013.

- International Labour Organization. Statistics on the informal economy. Report II for the 21st International Conference of Labour Statisticians ICLS/21/2023/II, Department of Statistics, International Labour Office, Geneva, 2023.
- International Labour Organization. ILOSTAT database: Employment by sex and occupation (EMP_TEMP_SEX_OCU_NB_A). <https://ilostat.ilo.org/data/>, 2026. Accessed 19 September 2026.
- Rafael La Porta and Andrei Shleifer. Informality and development. *Journal of Economic Perspectives*, 28(3):109–126, 2014.
- Shakked Noy and Whitney Zhang. Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192, 2023.
- Carlo Pizzinelli, Augustus Panton, Marina Mendes Tavares, Mauro Cazzaniga, and Longji Li. Labor market exposure to ai: Cross-country differences and distributional implications. IMF Working Paper WP/23/216, International Monetary Fund, 2023.
- U.S. Bureau of Labor Statistics. Occupational employment and wage statistics, may 2018 national estimates. <https://www.bls.gov/oes/tables.htm>, 2019. Accessed 19 September 2026.
- Michael Webb. The impact of artificial intelligence on the labor market. Technical report, Stanford University, 2020.
- World Bank. Living standards measurement study – integrated surveys on agriculture (lsms-isa). World Bank Microdata Library, 2024.

A Source of the task ratings

The task ratings come from the public repository that accompanies Althoff and Reichardt (2026), https://github.com/lukasalthoff/ai_labor_markets. The canonical file is `ai_capabilities/task_ai_capabilities.csv` at commit 4706d49 of 7 June 2026, downloaded on 18 September 2026, with SHA-256 checksum beginning `b56dcf23dd0a`.

Each occupation’s score is the task-weighted mean of its task ratings,

$$s_o = \frac{\sum_{t \in o} w_t a_t}{\sum_{t \in o} w_t},$$

where a_t is the binary rating of task t (`automatable_genai`) and w_t is its O*NET importance weight (`task_weight`). Applied to the canonical file, this formula reproduces all 974 scores in the published occupation file to within 10^{-16} . The eight alternative model and scenario files reported in Section 5.4 are fixed in the same way. The replication package records the commit, address, download date and full checksum of each file in `countries/althoff_provenance.json`.

B Survey variables

Table 14 lists the survey variables behind each harmonised concept. All come from the public LSMS-ISA release of each wave. In Nigeria the derived work-status variable (`s4aq21`) places a person on the family farm only when that farm is their sole work and its output is only or mainly for household use, on the answers to the destination questions in the last row.

Table 14: Survey variables used for each harmonised concept

Concept	Nigeria 2023	Tanzania 2020	Uganda 2019
Four-digit occupation	<code>s4aq40_code</code>	<code>hh_e30b_4a</code>	<code>h8q19b_fourDigit</code>
Wage employment	<code>s4aq51</code>	<code>hh_e03</code>	<code>s8q22</code>
Written contract	<code>s4aq56</code>	<code>hh_e42</code>	<code>s8q27</code>
Second formality marker	<code>s4aq57_1</code> (pension)	<code>hh_e44b</code> (tax withheld)	<code>s8q26</code> (PAYE)
Non-farm business owner	—	<code>hh_e05</code>	—
Farm work	<code>s4aq21</code>	<code>hh_e07</code>	—
Destination of farm output	<code>s4aq13a-c</code> , <code>s4aq19a-b</code>	<code>hh_e09</code>	—
Household weight	<code>wt_cross_wave5</code>	<code>y5_crossweight</code>	<code>wgt</code>

Nigeria’s variables are in the post-harvest labour module (`sect4a`) and household cover file (`secta`), Tanzania’s in `hh_sec_e1` and `hh_sec_a`, and Uganda’s in `gsec8` and `gsec1`. In Nigeria the destination questions refer to the last seven days (`s4aq13a-c`) or, for people with no farm work that week, to usual farm work (`s4aq19a-b`). A dash marks a concept the survey does not use for this purpose.

C Ethiopia

The Ethiopian Socioeconomic Panel Survey Wave 5 records occupation at the one-digit ISCO-88 major group, which cannot support the occupational argument of this paper. Ethiopia is therefore reported here and left out of the headline sample.

Two Ethiopian samples exist in the survey. The first holds 584 out-migrants, household members who had left and whose occupation the household reported. That sample is a poor

description of the Ethiopian workforce, with an agricultural share of 22.4 percent against a national figure above 60 percent, and it is left aside here. The resident wage sample is the better object. It holds 1,989 workers with a one-digit code, mapped to ISCO-08 exposure through the unit-group correspondence.

Resident Ethiopian wage workers average 0.216, weighted 0.190. Of them, 37.1 percent have no written contract, 45.9 percent on weights, and those workers average 0.129 against 0.267 for workers with a contract. The formal-informal ratio is close to the one found in the three headline countries. The sample covers wage jobs only and can say nothing about agriculture or own-account work, which is why it supplements the main analysis and does not enter it.

D Cross-country detail

Table 13 in the body reports one Atlas outcome and one ILOSTAT score vector. Four Atlas outcomes are available. They are the total exposed share, the share exposed through AI, the share exposed through learned content transformation, and the share exposed with no AI involved. Informality carries a large full-model coefficient on the first three and -0.006 with a standard error of 0.034 on the fourth, which cannot be told from zero. That is the basis for the claim in Section 6 that the Atlas informality result is a property of its AI classification.

Table 15: Atlas outcomes, full model

Outcome	African median	Informality alone	Informality full model	Agricultural self-emp.	Log GDP per head
Exposed, any route	0.242	-0.617 (0.077)	-0.248 (0.068)	-0.113 (0.074)	0.077 (0.015)
Exposed through AI	0.107	-0.482 (0.046)	-0.242 (0.049)	-0.093 (0.048)	0.045 (0.011)
Exposed, content transf.	0.068	-0.220 (0.029)	-0.087 (0.029)	-0.058 (0.028)	0.023 (0.006)
Exposed, no AI	0.132	-0.134 (0.034)	-0.006 (0.034)	-0.020 (0.032)	0.032 (0.007)

³² African economies; each model uses the countries reporting its covariates, 28 to 31. Heteroskedasticity-robust standard errors in parentheses. The agricultural self-employment and income coefficients are from the full model, which holds all three variables at once.

Table 13 reports the three predictors on their own scales, two of them shares and one a logarithm. Nothing in it ranks them. Table 16 puts them on a common footing. Standardised coefficients multiply each coefficient by the ratio of the predictor’s standard deviation to the outcome’s. Incremental R-squared is what the full model loses when that one predictor is

dropped. Both run on the 27 countries where both outcomes and all three predictors are present, with the same HC1 errors as Table 13.

The two measures disagree about which predictor leads. Under the ILOSTAT rebuild the agricultural share is first on both metrics. Under the Atlas measure income per head has the largest standardised coefficient while informality has the larger incremental R-squared, and the two are close enough that neither leads.

Table 16: Comparable predictor metrics for the two country measures

Predictor	Atlas, exposed through AI		ILOSTAT rebuild	
	Std. coef.	Incr. R^2	Std. coef.	Incr. R^2
Informal employment rate	−0.401	0.085	−0.182	0.018
Agricultural self-employment	−0.240	0.030	−0.492	0.124
Log GDP per head	0.406	0.077	0.301	0.043
Countries	27		27	
Full-model R^2	0.837		0.740	

Both columns use the full model of Table 13, which holds all three predictors at once, on the 27 countries where both outcomes and all three predictors are present. Standardised coefficients are the unstandardised coefficient times the predictor’s standard deviation over the outcome’s. Incremental R-squared is the full-model R-squared less the R-squared of the model without that predictor. Heteroskedasticity-robust standard errors as in Table 13. The unstandardised coefficients reproduce Table 13 on its own per-specification samples, 28 countries for the Atlas outcome and 27 for the ILOSTAT rebuild.

The ILOSTAT rebuild covers 45 African countries. Central African Republic, Libya and Mauritania report no data; Eritrea, South Sudan and South Africa report only ISCO-88; Algeria and Morocco carry no ISCO-08 series from 2015. Armed forces and unclassified workers are excluded and the shares rescaled over groups 1 to 9. Three score vectors give country medians of 0.123 (the mean over ISCO-08 unit codes), 0.088 (the survey-weighted worker scores) and 0.095 (the survey scores with own-use farmers excluded). The three rank countries almost identically, correlating at 0.98 or above, and the reported results hold under all three. Excluding the four countries with more than 20 percent of workers unclassified leaves the same pattern on 25 countries.

E Survey design

Table 17 gives the sampling design used for the standard errors reported throughout the paper.

Uganda’s design needs a caveat. The survey samples through enumeration areas, and no household file carries an enumeration-area code. The parish stands in. Of the 3,094

Table 17: Sampling design by country

Country	Primary sampling unit	PSUs	Strata	Fewest PSUs in a stratum
Nigeria	Survey cluster	731	6	103
Tanzania	Survey cluster	485	55	2
Uganda	Parish (2018 codes)	930	7	4
Ethiopia	Enumeration area	314	19	6

Strata are the survey’s own where supplied. For Uganda they are Kampala, other urban, and rural by region. Tanzania’s 545 booster households, added outside the 2014/15 frame, form one stratum per city with their enumeration area as the primary sampling unit; Tanga urban has a single such unit and joins Tanga rural. Ethiopia is the resident wage sample of Appendix C.

households with parish codes, 2,343 live in one of the 296 parishes covered by the community survey. The other 751, tracked movers and split-offs, sit alone or nearly alone in their parish and each counts as its own primary sampling unit, which understates clustering. Clustering Uganda more coarsely moves the standard error of its mean from 0.0017 at parish level to 0.0018 at subcounty, 0.0020 at county and 0.0025 at district. The standard errors of Uganda’s status-group means move by 0.0007 or less, some of them downward. The district figure is the honest upper bound.

Across the 35 group means, standard errors that allow for strata and clusters run from 0.96 to 1.55 times their household-clustered counterparts. The ratio is largest for groups whose members share villages, at 1.55 for Nigeria’s farmers, and close to one for clerical support and formal wage workers, who are scattered thinly across clusters.